

Note on the Galois Groups of Cubics

Let us prove the following lemma:

Lemma 1 *Suppose K is any field of characteristic other than 2 and $f \in K[x]$ is monic, irreducible, and has degree 3. Assume also that f has exactly 3 distinct roots in an algebraic closure of K , and let $D := -\text{Res}_{(3,2)}(f, f')$. Then $\text{Gal}_K(f)$ is isomorphic to A_3 or S_3 , according as D is the square of an element of K or not.*

We will prove the lemma from scratch, without using the Fundamental Theorem on Galois Correspondence. Before moving on, let us note that Lemma 1 breaks down in characteristic 2, owing to the fact that $1 = -1$ in such fields: Letting $\zeta_1, \zeta_2, \zeta_3$ denote the (distinct) roots of f in an algebraic closure of K , we have $(\zeta_1 - \zeta_2)(\zeta_1 - \zeta_3)(\zeta_2 - \zeta_3) = (\zeta_1 + \zeta_2)(\zeta_1 + \zeta_3)(\zeta_2 + \zeta_3)$, and the latter quantity lies in K since it is invariant under any permutation of the ζ_i . So, by the product formula for the discriminant, $-D = D = (\zeta_1 - \zeta_2)^2(\zeta_1 - \zeta_3)^2(\zeta_2 - \zeta_3)^2$ is always a square of an element in K when K has characteristic 2.

Proof of Lemma 1: Recall that $D = 0 \iff f$ has a root in common with f' in the algebraic closure of K . Since f is irreducible in $K[x]$, and the ζ_i are distinct, we must have $D \neq 0$. (An earlier hand-out relating the Sylvester resultant and the discriminant proves the equivalence of $D = 0$ with having roots of multiplicity ≥ 2 over $\mathbb{C}$, and the proof there in fact extends transparently to arbitrary algebraically closed fields.)

Now note that $\text{Gal}_K(f)$ has cardinality ≤ 6 (since it is a group acting faithfully on 3 roots) and cardinality at least 3. The lower bound follows from the fact that $K(\zeta_1)$ is isomorphic to $K(\zeta_1)$, $K(\zeta_2)$, and $K(\zeta_3)$, and these 3 isomorphisms extend to (at least) 3 distinct automorphisms of $K(\zeta_1, \zeta_2, \zeta_3)$. To see why, let us separate into two cases: (a) f splits completely over $K(\zeta_1)$ and (b) f does *not* split completely over $K(\zeta_1)$. In Case (a), the aforementioned isomorphisms are already automorphisms of $K(\zeta_1, \zeta_2, \zeta_3) = K(\zeta_1)$, and ζ_1 has 3 distinct images under these automorphisms. In Case (b), f must factor as $(x - \zeta_1)g(x)$ over the field $K(\zeta_1)$, for some irreducible quadratic $g \in K(\zeta_1)[x]$. So then, $K(\zeta_1, \zeta_2, \zeta_3)$ is clearly isomorphic to each of $K(\zeta_1, \zeta_2)$, $K(\zeta_1, \zeta_3)$, and $K(\zeta_2, \zeta_3)$; and we obtain our desired extensions by simply mapping the pair (ζ_1, ζ_2) to any pair of the form (ζ_i, ζ_j) with $i \neq j$. (This in turn forces the image of ζ_3 to be ζ_k where $\{k\} = \{1, 2, 3\} \setminus \{i, j\}$.) There are thus 6 automorphisms possible, so we obtain that $\text{Gal}_K(f)$ has cardinality 6 in Case (b).

Now we make the connection to the discriminant: From our earlier work on the Sylvester resultant, and the fact that f is monic, we have that $D = (\zeta_1 - \zeta_2)^2(\zeta_1 - \zeta_3)^2(\zeta_2 - \zeta_3)^2$. So if D is a square in K we must have that $r := (\zeta_1 - \zeta_2)(\zeta_1 - \zeta_3)(\zeta_2 - \zeta_3) \in K$. If $\text{Gal}_K(f) \cong S_3$ then it would contain a transposition, but such an element would change the sign of r . This is impossible, since r needs to be fixed by $\text{Gal}_K(f)$. So D a square in K indeed implies that $\text{Gal}_K(f) \cong A_3$.

To conclude, we need only show that D *not* a square in K implies that f does not split completely over $K(\zeta_1)$. So observe first that D not a square in K implies that $[K(\sqrt{D}) : K] = 2$. We already know that $[K(\zeta_1) : K] = 3$. So, by the multiplicativity of field extension degree in towers of extensions, $6 \mid [K(\zeta_1, \zeta_2, \zeta_3) : K]$ and thus $K(\zeta_1, \zeta_2, \zeta_3) \neq K(\zeta_1)$. In particular, from our earlier observations, we see that $\text{Gal}_K(f) \cong S_3$. ■

It also happens to be the case that, for f as above, and any root ζ_1 of f in an algebraic closure of K , factoring f as $(x - \zeta_1)g(x)$ for some quadratic $g \in K(\zeta_1)(x)$ implies that $-\text{Res}_{(3,2)}(f, f') = g^2 \text{Res}_{(2,1)}(g, g')$. So we in fact have that $K(\zeta_1, \sqrt{D}) = K(\zeta_1, \zeta_2 - \zeta_3)$. Thanks to our observations on the underlying field extension degrees above, we thus obtain that the splitting field of f over K is exactly $K(\zeta_1, \sqrt{D})$.